9 **(a)** Radioactive isotopes that emit ionising radiation are used in hospitals.

(i) State and explain **two** safety precautions necessary for the use of these isotopes in medical procedures.

safety procedure 1 .....

explanation .....

.....

safety procedure 2 .....

explanation .....

.....

[2]

(ii) Give **two** reasons why alpha-emitters are **not** used as radioactive tracers inside the body.

1 .....

.....

2 .....

.....

[2]

(b) Sodium-24 is an isotope of sodium (Na) that has a proton number of 11 and a nucleon number of 24.

Sodium-24 decays by emission of a beta-particle to form an isotope of magnesium (Mg).

Use nuclide notation to write down the nuclide equation for this decay.

[3]

[Total: 7]

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